

Building a Unified Multicloud Strategy for Goldman Sachs

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Table of Contents

Section 1: Cloud Governance Problem Statement	3
Section 2: Unified Multicloud Strategy and Architecture	4
Section 3: Identity and Access Governance	7
Section 4: Data Protection and Infrastructure Security Controls	9
Section 5: Cost, Vendor Risk, and Accountability	12
Section 6: Multicloud Strategy and Provider Governance	14
Section 7: AI Governance and Responsible Use	16
Conclusion	17
Works Cited	18

Table of Figures

Figure 1: Goldman Sachs Multicloud Architecture Diagram	4
Figure 2: Architecture Tradeoff Evaluation	5
Figure 3: Accepted Risk Summary	6
Figure 4: Role List	7
Figure 5: Access Model Matrix	8
Figure 6: Data Classification Table	10
Figure 7: Security Controls Applied to the Multicloud Architecture	11
Figure 8: Cost Drivers and Ownership	12
Figure 9: Resource Tagging Example — AWS Redshift Cluster	13

Section 1: Cloud Governance Problem Statement

Goldman Sachs is a leading global financial institution that relies on cloud platforms to support trading, data processing, analytics, and innovation. AWS supports core trading systems and large-scale data processing, while Google Cloud supports advanced analytics and machine-learning workloads ("Goldman Sachs Partners AWS"; Fluence).

Operating across multiple cloud providers creates governance challenges because separate tools, policies, and management practices can reduce visibility and increase operational complexity. These challenges are particularly significant because Goldman Sachs operates under strict regulatory requirements. SEC Rule 17a-4, FINRA Rule 4511, and Regulation S-P require consistent recordkeeping, audit trails, access controls, and data protection across cloud environments (FINRA; Keepit).

The primary governance challenge is maintaining consistent oversight across AWS and Google Cloud. Differences in logging, data retention, identity management, monitoring, and cost-accountability practices can create governance gaps when cloud environments are managed independently.

Three major governance risks emerge from this multicloud model:

1. Inconsistent logging and retention controls may result in compliance failures.
2. Fragmented monitoring capabilities can limit the firm's ability to detect and respond to security incidents.
3. Independently managed cloud resources can create ownership and cost-accountability challenges.

To address these risks, Goldman Sachs must answer two leadership-level governance questions:

1. Who holds decision rights and accountability for enforcing cloud-governance policies across AWS and Google Cloud?

2. Under what conditions can a business unit adopt a new cloud platform or service, and who has authority to approve exceptions to governance policies?

Section 2: Unified Multicloud Strategy and Architecture

Architecture Diagram

The architecture separates AWS trading systems from Google Cloud analytics through a governed cross-cloud connection. *(See Figure 1: Goldman Sachs Multicloud Architecture Diagram below.)*

Users / Clients : Traders, Analysts, Internal Apps	
Public Entry Point: AWS CloudFront + API Gateway	
Application / Service Layer: AWS EC2	Application / Service Layer: GCP Vertex AI
Data Layer: AWS S3, RDS, Redshift	Data Layer: GCP BigQuery (Analytics) Store)
AWS <-> GCP Cross-Cloud Connection Trading records & market datasets IAM + Data Steward approval	
Identity & Access Management: AWS IAM + GCP IAM	
Logging & Monitoring: AWS CloudWatch and GCP Cloud Logging	
External / SaaS Dependencies: Bloomberg Data Feed · FINRA · SEC Reporting Endpoints	

Figure 1: Goldman Sachs Multicloud Architecture Diagram

Architecture Pattern

Goldman Sachs operates a multicloud architecture that uses AWS for core trading systems, transaction processing, storage, security, and monitoring workloads, while Google Cloud supports analytics and AI capabilities through BigQuery and Vertex AI.

AWS-GCP Dependency

Trading records and market datasets are shared between AWS and Google Cloud through a governed, encrypted connection controlled by IAM policies, Data Steward approval, and audit logging.

Resilience Statement

Core trading operations remain available within AWS if Google Cloud analytics services become unavailable. While analytics and AI capabilities may be affected, trade execution and regulatory record retention continue operating within AWS..

Architecture Tradeoff Evaluation

This evaluation compares the proposed architecture across key technical, operational, and governance criteria.

[*\(See Figure 2: Architecture Tradeoff Evaluation below.\)*](#)

Dimension	Score	Rationale
Security	4 / 5	AWS and GCP both offer encryption, IAM, and compliance tooling. Having a unified policy is a present gap with security.
Cost Predictability	3 / 5	Managed services improve cost stability, but multicloud licensing, inter-provider fees, and separate tools reduce cost predictability.
Resilience	4 / 5	Multi-AZ deployment on AWS and GCP's regional redundancy provide strong uptime. Single-vendor outage risk is mitigated by the hybrid model.
Staff Skills	4 / 5	Goldman Sachs employs specialized cloud engineers. AWS depth is strong; GCP ML expertise exists but is narrower.
Vendor Lock-in	3 / 5	AWS-dependent core services (Redshift, ECS) and added GCP BigQuery layer limit portability without major refactoring.
Time to Launch	3 / 5	Managed services speed deployment, but multicloud governance, compliance requirements, and integration complexity slow initial rollout.

Figure 2: Architecture Tradeoff Evaluation

Weakest Dimension: Cost Predictability: Cost predictability remains the weakest dimension of the architecture. Operating across AWS and Google Cloud introduces separate billing models, cloud-management tools, and inter-cloud transfer charges that are more difficult to forecast than in a single-cloud environment.

Accepted Risk

This section identifies the risk accepted by the organization and the controls used to manage its impact.

The accepted risk reflects a deliberate business decision in which the expected benefits outweigh the remaining risk after mitigation controls are applied.

[*\(See Figure 3: Accepted Risk Summary below.\)*](#)

Risk	Goldman Sachs accepts the operational dependency created by sharing trading records and market datasets between AWS and Google Cloud for analytics and AI workloads.
Why Accepted	The architecture enables advanced analytics and AI capabilities while maintaining core trading systems and regulated data within AWS.
Mitigation if Materialized	Core trading workloads operate independently within AWS if Google Cloud analytics services become unavailable. Analytics and AI capabilities may be degraded, but trade execution, transaction processing, and regulatory record retention continue operating within AWS. Cross-cloud dependencies are limited to analytics workloads, and fallback procedures are maintained for analytics interruptions.

Figure 3: Accepted Risk Summary

Risk Mitigation vs. Risk Transfer

Goldman Sachs mitigates multicloud dependency risk through architectural controls rather than relying on cloud-provider service-level agreements (SLAs). While SLAs may provide compensation after an outage, they do not reduce the likelihood or impact of the outage itself. Risk mitigation is achieved by limiting critical runtime dependencies on Google Cloud, maintaining fallback procedures, and ensuring core trading operations can function independently within AWS.

Section 3: Identity and Access Governance

Goldman Sachs uses identity and access governance to protect regulated data, enforce least-privilege access, and maintain consistent security controls across AWS and Google Cloud. Clearly defined roles, approval processes, and accountability structures ensure that access is granted according to business need while supporting regulatory and audit requirements.

Role List

The following roles define access governance responsibilities across the multicloud environment

[\(See Figure 4: Role list below.\)](#)

Role	Access Type	Responsibility
Cloud Platform Engineer	Permanent	Manages AWS/GCP infrastructure, IAM, networking, and AWS-GCP connectivity.
Application Owner	Permanent	Manages trading systems and analytics integrations.
Data Steward	Permanent	Governs datasets and approves AWS-GCP data sharing.
Security Reviewer	Permanent	Reviews privileged access, IAM changes, and audit logs.
Finance Lead	Permanent (View-Only)	Monitors cloud and transfer costs.
External Auditor	Time-Limited	Reviews audit evidence and compliance logs.
Vendor Consultant	Time-Limited	Supports approved AWS/GCP configurations.

Figure 4: Role list table

Access Model

The access model defines how permissions are assigned across AWS and Google Cloud based on operational responsibilities and business needs. Permanent access is reserved for roles that require ongoing involvement in managing infrastructure, applications, security, or compliance, while time-limited access is granted only for approved audit, consulting, or support activities. This approach supports least-privilege principles while reducing the risk of unnecessary long-term access.

[\(See Figure 5: Access Model Matrix below.\)](#)

Role	Trading Apps	IAM / Net	Data Stores	Cross-Cloud Transfer / Integration	Logs / Reports	Approval
Cloud Platform Engineer	Modify/Admin	Admin	View/Support	administer connection and transfers	View	Co-approve privileged changes
Application Owner	Modify	None	View approved app data	Modify approved integrations	View	Request privileged access
Data Steward	None	None	Modify/Approve data access	Approve shared datasets	View	Approve data sharing
Security Reviewer	View	Approve/View	View	Approve/review permission changes	View/Review	Approve privileged access
Finance Lead	None	None	None	View transfer-cost reporting	View cost reports	None
External Auditor	View	View	View approved records	View approved evidence	View	None
Vendor Consultant	Time-limited modify on approved systems	None unless approved	None unless scoped	Time-limited changes to approved integrations	Limited view	None

Figure 5: Access Model Matrix

Time-Limited Access: Vendor Consultant and External Auditor access expires automatically at the end of the approved engagement or audit period.

Dual Approval : Changes to administrative access or AWS-GCP data permissions require approval from the Business Owner or Manager and validation by the Security Reviewer. The Cloud Platform Engineer implements the approved change but does not serve as an approver.

Governance Answers

Privileged Access Workflow:

Step 1: The Application Owner submits an access request with business justification.

Step 2: The Business Owner or Manager reviews and approves the business need.

Step 3: The Security Reviewer validates least-privilege requirements and evaluates cross-cloud risk.

Step 4: The Cloud Platform Engineer implements the approved access change with MFA enforced.

Quarterly Evidence:

Leadership reviews quarterly governance reports focused on:

- New privileged access granted during the review period
- Cross-cloud permission changes
- Policy exceptions and remediation items
- MFA violations or access-control deviations
- Verification that time-limited access expired as expected

Supporting evidence is collected through IAM access reports, AWS CloudTrail, and GCP Cloud Logging summaries of administrative activity and cross-cloud access changes.

Section 4: Data Protection and Infrastructure Security Controls

Goldman Sachs protects regulated financial information through data classification, encryption, access controls, network security boundaries, and audit logging. These controls are applied across AWS and Google Cloud to support regulatory compliance, prevent unauthorized access, and protect sensitive financial data throughout its lifecycle.

Data Classification Table

The table below categorizes the sensitivity of data handled within the Goldman Sachs multicloud environment and identifies the regulatory or business driver behind each classification.

(See Figure 6: Data Classification Table below.)

Data Item	Classification	Justification
Trade transaction records	Restricted	Contains detailed financial transaction data describing identifiable client trades; governed by SEC/FINRA audit, compliance, and recordkeeping requirements
Client PII (account + identity data)	Restricted	Includes personal and financial identifiers describing individual clients; protected under Regulation S-P with severe breach impact.
Market analytics datasets	Confidential	Aggregated data used in GCP analytics; does not directly identify clients but is business-sensitive.
System logs (CloudTrail, GCP logs)	Confidential	Records access and activity describing internal users and administrators; exposure could reveal system behavior or security gaps.
Application configuration files	Internal	Contains system settings and service configurations describing infrastructure behavior; does not directly identify users and has limited exposure risk.
Public financial research reports	Public	Publicly released financial reports describing market insights for external audiences; intended for sharing with no risk if exposed.
IAM roles and access policies	Confidential	Defines permissions and access boundaries describing user and system privileges; exposure could enable privilege escalation and security compromise.

Figure 6: Data Classification Table

Security Controls Applied to the Multicloud Architecture

The data classification table defines the sensitivity of information handled within the Goldman Sachs environment. The architecture below shows the security controls used to protect that data across AWS and Google Cloud.

(See Figure 7: Security Controls Applied to the Multicloud Architecture below.)

Users / Clients : Traders, Analysts, Internal Apps TLS 1.2+	
Public Entry Point: AWS CloudFront + API Gateway • PUBLIC-PRIVATE BOUNDARY • TLS Termination • Web Application Firewall (WAF)	
Application / Service Layer: AWS EC2 • Security Group: API Gateway Only • No Public Inbound Access • TLS Encryption	Application / Service Layer: GCP Vertex AI (Analytics & AI) • Firewall: Approved AWS Traffic Only • Least-Privilege Access • TLS Encryption
Data Layer (AWS S3, RDS, Redshift) • AWS KMS (CMK – Goldman-Owned Keys) • IAM-Controlled Access • AWS Config Monitoring	Data Layer: GCP BigQuery (Analytics) Store) • Customer-Managed Encryption Keys (CMEK) • IAM-Restricted Access
AWS <-> GCP Cross-Cloud Connection • Trading Records & Market Datasets • IAM + Data Steward Approval TLS Encrypted Transmission	
Identity & Access Management: AWS IAM + GCP IAM • Federated Identity (SSO) • MFA Required • Least Privilege Enforced	
Logging & Monitoring: AWS CloudWatch and GCP Cloud Logging • AWS CloudTrail • GCP Cloud Logging	
External / SaaS Dependencies: Bloomberg Data Feed · FINRA · SEC Reporting Endpoints • TLS-Secured Connections • Audit Logging Enabled • Approved Data Exchanges Only	

Figure 7: Security Controls Applied to the Multicloud Architecture

Potential Data-Exposure Risk

An AWS S3 bucket containing restricted client trade records could become publicly accessible because of a configuration error. AWS Config Rules continuously monitor bucket settings and identify unauthorized changes before sensitive information is exposed.

How Security Controls Work Together

TLS protects data moving between AWS and Google Cloud, while IAM policies, network boundaries, and CMK/CMEK encryption protect stored data in S3, Redshift, and BigQuery. These controls work together to protect regulated financial information and support compliance requirements.

Section 5: Cost, Vendor Risk, and Accountability

Cost Drivers & Ownership

Managing a multicloud environment requires accountability for cloud spending, vendor dependencies, and regulatory obligations. Goldman Sachs addresses these risks through cost ownership, resource tagging, vendor oversight, and compliance governance.

(See Figure 8: Cost Drivers and Ownership below.)

Cost Driver	Owner (role-level)
AWS EC2 compute for core trading applications	Trading Systems Product Owner
Redshift + RDS data layer (storage and processing)	Head of Data Platforms
GCP Vertex AI / BigQuery analytics workloads	Analytics & ML Product Owner
Cross-cloud data transfer (AWS & GCP)	FinOps lead
Bloomberg data feed licensing & throughput	Market data services owner

Figure 8: Cost Drivers and Ownership

Resource Tagging

To support cost accountability, governance, and compliance reporting, Goldman Sachs applies a standardized tagging strategy across cloud resources.

Required tags: owner, environment, cost_center, compliance_flag.

Example: AWS Redshift Cluster

(See Figure 9: Resource Tagging Example — AWS Redshift Cluster below.)

Tag	Example resource: AWS Redshift cluster (The Data Layer)
owner	Head of Data Platforms
environment	Production
cost_center	Trading_Operations
compliance_flag	SEC_FINRA_regulated

Figure 9: Resource Tagging Example — AWS Redshift Cluster

Vendor Dependencies

- AWS: Goldman Sachs depends on AWS for core trading operations. The principal lock-in risk is not moving compute or databases, but migrating the connected data, identity, monitoring, and application dependencies built around them. The most important contract clause is exit and portability.
- Bloomberg Data Feed: Trading and analytics processes depend on Bloomberg's proprietary market data. Replacing Bloomberg would require changes to ingestion pipelines and trading applications. The most important contract clause is data processing terms.

Compliance Responsibilities

Trade execution and order data stored in AWS Redshift and S3 are subject to SEC Rule 17a-4 and FINRA record-retention requirements.

Evidence: AWS CloudTrail logs and retained trade records must be available during regulatory audits.

Shared Responsibility: AWS manages the underlying infrastructure, while Goldman Sachs remains responsible for data classification, retention policies, access controls, audit evidence, and regulatory compliance.

Cost, Vendor, and Compliance Overlap

The Redshift data layer is a major cost driver, a significant AWS dependency, and a repository for regulated trade records. To reduce compliance risk, retention policies are explicitly configured, regularly validated, and owned by the Head of Data Platforms. Compliance is verified through CloudTrail evidence, retention reviews, and configuration validation rather than assumed from AWS service commitments.

Section 6: Multicloud Strategy and Provider Governance

Multicloud Position

Goldman Sachs operates a multicloud environment that uses AWS and Google Cloud for different business objectives. AWS supports core trading systems, storage, security controls, and monitoring workloads, while Google Cloud hosts analytics, artificial intelligence, and machine-learning capabilities through BigQuery and Vertex AI.

Analytics, machine-learning, and AI workloads remain on Google Cloud because of data gravity and a genuine best-of-breed need rather than simple preference. The datasets used to train models already reside within BigQuery, and exporting large volumes of data to AWS would require significant time and cost. Existing analytics workflows are also built around Vertex AI, meaning migration would require redesigning and rebuilding established pipelines.

Driver-Owner-Exit Test

1. Driver: Advanced analytics and machine-learning capabilities supported by BigQuery and Vertex AI.
2. Owner: Analytics & ML Product Owner.
3. Exit Cost: Migrating to AWS would require moving large datasets and rebuilding analytics pipelines and supporting services.

Identity Governance

The primary governance challenge within the multicloud environment is maintaining identity consistency across AWS and Google Cloud. Rather than managing identities separately, Goldman Sachs will centralize identity through a federated identity model using a single enterprise identity provider and single sign-on (SSO). AWS and Google Cloud will retain their native permission structures, but authentication and user lifecycle management will be governed centrally.

Accountable Owner: Director of Identity and Access Management.

Identity as a Governance Foundation

Identity should be centralized before other governance initiatives because cost reporting, logging, monitoring, and compliance activities all depend on knowing who accessed a resource and when. Once identity is centralized, logging, monitoring, compliance reporting, and cost governance can be implemented more effectively across both cloud environments.

Section 7: AI Governance and Responsible Use

AI Opportunity

Goldman Sachs can leverage BigQuery and Vertex AI to support analytics, pattern detection, forecasting, and decision support across trading and business operations. These capabilities can improve efficiency, accelerate data analysis, and enhance business insights.

Data Governance for AI

AI systems inherit the risks, classifications, and regulatory obligations of the data used to train and operate them. Therefore, all data used in AI workloads must follow the same governance requirements applied throughout the AWS and Google Cloud environment. Datasets that have not been approved for analytics use, are subject to legal restrictions, or cannot satisfy regulatory requirements are considered out of scope for AI initiatives.

AI Review and Approval

New AI use cases should be reviewed before implementation to ensure that the intended business objective, data sources, security requirements, and compliance obligations are clearly understood and approved.

Accountable Owner: Analytics & ML Product Owner.

Human Decision Authority

AI may provide recommendations, predictions, and analytical insights, but final business decisions remain the responsibility of authorized personnel. Trading decisions, regulatory determinations, risk acceptance decisions, and compliance actions require human review and approval.

Cost and Oversight

AI workloads can generate significant compute, storage, and model-training costs. Usage and cloud consumption should be monitored regularly to ensure that AI initiatives continue to deliver business value while remaining aligned with governance, security, and compliance requirements.

Conclusion

Goldman Sachs' multicloud strategy balances regulatory control, security, and innovation across AWS and Google Cloud. Clear governance structures, defined ownership, strong security controls, and centralized identity management support consistent operations across both environments.

By implementing these governance practices, Goldman Sachs can reduce risk, maintain compliance, manage multicloud complexity, and support the responsible use of analytics and AI capabilities.

Works Cited

FINRA. “Regulatory Considerations for Cloud Computing.” *FINRA*, 2026,
<https://www.finra.org/rules-guidance/key-topics/fintech/report/cloud-computing/regulatory-considerations>
. Accessed 21 July 2026.

Fluence. “5 Enterprise Multi-Cloud Examples That Slash Costs and Boost Resilience”. *Fluence Network*,
2025, <https://www.fluence.network/blog/multi-cloud-examples-5-case-studies/>. Accessed 21 July 2026.

Fintechfutures. “Goldman Sachs Partners AWS for Cloud Data Management and Analytics Solution.”
Fintech Futures, 2025,
<https://www.fintechfutures.com/cloud-services/goldman-sachs-partners-aws-for-cloud-data-management-and-analytics-solution>. Accessed 21 July 2026.

Keepit. “What FINRA and SEC Compliance Requires and How Backup Solutions Can Help.” *Keepit*,
2024, <https://www.keepit.com/blog/finra-sec/>. Accessed 21 July 2026.

